Selective Language Damage with Parameter-Decomposition Subcomponents

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I study selective language damage from editing individual adVersarial Parameter Decomposition (VPD) subcomponents in a four-layer, 67M-parameter model, measuring the effects on a target language and its close relative. A fixed rule uses target and English text to select one subcomponent per target. Six edits across three pairs (German and Dutch, Spanish and Portuguese, French and Italian) raise target cross-entropy (CE) by 0.5 or 1 nat/token on validation text. I compare them with rank-one LoRA, full-matrix fine-tuning and neuron edits calibrated the same way.

On 58 European Parliament session dates evaluated once after the edits were fixed, the subcomponent edits raised the related language's CE by at most 0.0053 nat/token. All 18 simultaneous 95% intervals for the differences from the comparators lay below zero. The subcomponent edits therefore did less related-language damage at matched validation effects. A post hoc check found achieved target effects to agree within ±0.1 nat/token in 15 of 18 comparisons. The learned updates preserved English output better.

Exploratory comparisons on previously evaluated development dates generally narrowed the gaps to learned updates trained for 2,000 steps, while retaining the subcomponent edits' advantage at the tested budgets. The subcomponent edits also did less related-language damage than a whitened rank-one selector. The result concerns selective fixed edits in this model; the dictionary's contribution, the mechanism and generalization of the selection rule remain unresolved.

1 Introduction

A model edit that changes one behaviour usually changes others too. Model-editing work evaluates *locality*: inputs outside an edit's scope should keep their predictions (Yao et al. 2023, sec. 2). ROME, for example, scores an edited fact by whether the model retains facts about nearby subjects (Meng et al. 2022, sec. 3.3).

I use deliberate language damage as a controlled test of edit selectivity: each edit is calibrated to the same target-language damage on validation text, then evaluated for its effects on another language. A demanding neighbour for a language edit is a closely related language. Zeroing the parameters or neurons most important for one language also degrades its relatives: Spanish removal raises Italian and Portuguese perplexity (Zhang et al. 2024, Table 11), and deactivating French neurons also affects Spanish (Tang et al. 2024, sec. 3.3.1).

Parameter decompositions may allow narrower edits. adVersarial Parameter Decomposition (VPD)[1] approximates each weight matrix of a model by a sum of rank-one *subcomponents*, trained so that few are needed on any input (Bushnaq et al. 2026, sec. 2). It follows earlier attribution-based and stochastic decompositions (Braun et al. 2025; Bushnaq et al. 2025). If a subcomponent is specific to one language, an edit along it, $W \leftarrow W + \alpha A$, might change that language and spare its relatives. This study uses Goodfire's four-layer, 67M-parameter decoder-only model decomposed in the VPD report.

This study is a small follow-up inspired by Lucius Bushnaq's LessWrong post, "Exploration: fine-tuning with parameter decomposition". Bushnaq trains the scalar prefactors of selected subcomponents of the same model to suppress German. One subcomponent, chosen by its interpretation label, reaches chance-level German with little English damage and spares French and Spanish, while also damaging Italian (Bushnaq 2026). His comparison uses matched training budgets and LoRA baselines, with varying achieved effects.

I extend Bushnaq's scalar-edit comparison to three language pairs with a fixed rule for selecting one subcomponent per target from target and English text. I match target effects on validation text and add full-matrix fine-tuning and neuron baselines.

[1] A separate experiment in the VPD report tests manual changes to a subcomponent's write vector using an unembedding direction. A LoRA update of the same matrix trained on 947 examples does better on both of its KL measures (Bushnaq et al. 2026, sec. 6).

The pairs are German (de) and Dutch (nl), Spanish (es) and Portuguese (pt), and French (fr) and Italian (it). In each pair the related language is *protected*: it is evaluated after selection and training. I calibrate each edit to raise target cross-entropy (CE) by $\tau = 0.5$ or 1 nat/token on validation text.

The outcome is the CE rise of the protected language on European Parliament session text (Koehn 2005). After fixing the six edits and comparators, I evaluate them once on 58 new session dates to test whether the difference in related-language damage carries over to new text.

2 Setup

The method details give data sizes, selection and training recipes, and interval construction.

Model and subcomponents. The base model W_0 is the four-layer, 67M-parameter decoder-only model decomposed in the VPD report. It was trained on a subset of the Pile (Bushnaq et al. 2026, sec. 3.1; Gao et al. 2020). Edits act on its 24 weight matrices: in each layer, the query, key, value and output projections of attention and the two MLP matrices. The decomposition supplies 38,912 rank-one subcomponents in all, each of the form $A_c = u_c v_c^\top$. Subcomponents are named by layer, matrix and index: `0.attn.o:268` is subcomponent 268 of the attention output projection of layer 0.

The edit. An edit along subcomponent c of matrix m is $W_m \leftarrow W_m + \alpha A_c$, with the subcomponent not normalized. In VPD every subcomponent carries a mask coefficient that equals 1 in the unedited model. The edit moves it to $1 + \alpha$. We write $\alpha = s \cdot a$, with a sign $s = \pm 1$ and a dose $a > 0$. With $s = -1$, as for every subcomponent selected here, the coefficient is $1 - a$: a dose below 1 shrinks the subcomponent's contribution, a dose of 1 removes it, and a dose above 1 reverses its sign. All other subcomponents and the decomposition residual stay fixed. The edit norm is $|\alpha| \, \|A_c\|_F$.

Text and effects. Every Europarl set has one window per session date, parallel across languages. The *CE rise* of an edit is measured in nat/token. For each window, I subtract the base model W_0's mean next-token CE from the edited model's, then average these differences over windows. Each target has three overlapping *draws* of its Europarl training windows. The English *retain pool* mixes Europarl English training text, WikiText-2 validation text (Merity et al. 2017) and English-filtered Pile validation text.

2.1 Selecting the subcomponent

The rule scores each subcomponent by its local effect on target CE relative to its local effect on English output. Let g_c be the derivative of the mean target CE on the draw with respect to the subcomponent's mask coefficient. Let q_c be the curvature of the English output along the same coordinate, estimated on the retain pool with labels sampled from W_0. This estimates the directional Fisher information of the model's predictive distribution. The rule ranks subcomponents by the *gradient-to-retain-curvature score*

$$\frac{g_c^2}{q_c + \varepsilon \|A_c\|^2},$$

where a small floor ε keeps subcomponents with almost no measured curvature from dominating.

The rule then tests a shortlist of the best-ranked subcomponents on the target training draw. For each subcomponent, with sign $s = \operatorname{sign}(g_c)$, it raises the dose step by step up to 2. It accepts the first subcomponent whose target CE rise reaches 1 nat/token. If none does, it accepts the first that reached 0.5, and otherwise it finds no feasible subcomponent.

The rule chose the same subcomponent in all three draws of each target: `0.attn.o:268` for German, `0.attn.o:676` for Spanish and `0.attn.o:640` for French. They all lie in the attention output projection of the first layer and have sign -1.

2.2 Comparators

The comparators test other ways to produce the target effect, with different choices of which weights may change. Like the selection rule, they use only target and English text for selection and training.

- **Rank-one LoRA** (Hu et al. 2022) learns both update vectors in layer 0's attention output matrix, matching the subcomponent edit's matrix and rank while learning its own direction.
- **Full-matrix fine-tuning** updates all 24 weight matrices, allowing a broader learned edit.
- **Neuron edits** use the top k MLP neurons ranked by how much more often they are active on the target draw than on the English retain pool. The edit scales their output weights down, with k and the scale calibrated to τ, in the spirit of language-specific neurons (Tang et al. 2024; Kojima et al. 2024). This localizes the edit through neuron activity rather than a decomposition of the weights.
- **The randomized-dictionary control** applies the same selection rule to a randomized dictionary. In each matrix, one random signed permutation transforms the output coordinates of every subcomponent, and another transforms the input coordinates. This preserves their norms and pairwise angles but removes their alignment with W_0.

The two learned updates are trained separately for each draw, for 200 steps. They minimize a per-window hinge penalty for target CE rises below 1 nat/token, plus the KL divergence from W_0 on the English retain pool with weight 1. The final update is scaled down to reach τ.

Additional comparisons vary the choice of a single subcomponent within the same dictionary. They use a separate multiplicity correction and receive no confirmatory verdicts (Section 3.6).

2.3 Calibration and the six edits

An edit could spare the protected language simply because it has little effect on the target. To compare methods at a common target effect, I calibrate every method on target Europarl validation text. A short search over its dose sets the target CE rise to τ. An edit is *matched* if its validation effect lies within 0.02 nat/token of τ. Only matched edits enter a contrast. The effects achieved on the readout dates can differ and are reported with the results.

Table 1: The six VPD subcomponent interventions, each shared by the three draws: dose a, the subcomponent's mask coefficient $1 - a$ after the edit, edit norm $|\alpha| \cdot \|A\|$ and the achieved CE rises in nat/token. C: confirmation. D: development.

Pair	Subcomponent	τ	Dose a	Coefficient	Edit norm	Target rise, D	Target rise, C	Protected rise, C
de → nl	0.attn.o:268	0.5	0.681	0.319	0.29	0.49	0.48	0.0030
		1	0.945	0.055	0.40	1.00	0.98	0.0053
es → pt	0.attn.o:676	0.5	1.026	−0.026	0.34	0.53	0.52	0.0003
		1	1.388	−0.388	0.47	1.05	1.02	0.0030
fr → it	0.attn.o:640	0.5	0.762	0.238	0.31	0.53	0.50	0.0008
		1	1.039	−0.039	0.42	1.06	1.01	0.0017

The three draws yield the same six fixed scaled edits, one per target and τ: three reduce the selected contribution and three reverse its sign; none removes it completely (Table 1).

2.4 What was fixed, and what was known

Selection and training used only target and English text. Earlier research had used protected-language development text and identified the German subcomponent, so the selection history was not blind (Section A.3).

I fixed the edits, comparators and analysis before evaluating them once on 58 further session dates. Those dates were not used for selection, calibration or re-matching. This tests the six fixed edits on new text in the same languages. Evaluating the selection procedure itself remains an open question.

2.5 Readouts, eligibility and intervals

Guards. English test sets determine which matched edits are usable: Europarl, WikiText-2 and Pile test text on the development readout, and English text from the confirmatory dates on the confirmatory readout. An edit is unusable if any guard's CE rises by more than 0.1 nat/token. On the confirmatory dates it must also have been usable on development. This is an eligibility rule fixed in advance. Two usable edits can still differ greatly in how much they change English (Section 3.4).

Contrasts. For each protected-language session date, a contrast takes the CE rise of the subcomponent edit minus that of the comparator, averaged over the draws in which both are matched and usable. Its estimate is the mean over dates. Negative values mean less protected-language damage for the subcomponent edit. The *claim family* comprises four comparators × three pairs × two values of τ, or 24 planned contrasts.

Intervals. Every confidence interval in this paper is a simultaneous 95% interval from a bootstrap over session dates (Efron 1979), with one resample shared by every contrast of a family and the max-|t| critical value of a simultaneous band (Montiel Olea and Plagborg-Møller 2019, supplement, Algorithm 3). The intervals describe uncertainty across session dates for these fixed edits. The bootstrap varies dates while holding the draws, training seeds, subcomponents, language pairs and model fixed.

Confirmatory verdicts. Each claim contrast with a development interval receives one verdict on the confirmatory readout, according to a rule table fixed before the set was opened. It is "reproduced on confirm" if its interval excludes zero with the development sign and no used draw has a target CE rise outside $\tau \pm 0.1$. A rise outside this band is the *target flag*.

Post hoc analyses. Two analyses were chosen after the results were known, outside the protocol. The frozen verdicts remain those of the original analysis. The first asks whether the target effects achieved on the readout dates were comparable (Section 3.3). The second counts how often the learned updates move English less (Section 3.4).

3 Results

3.1 Target change against protected damage

Figure 1 places the intended target effect next to the unwanted protected-language damage. All matched edits passed the English guards, and on the confirmatory dates no edit raised the target flag. At both values of τ and for all three pairs, the subcomponent edits have the lowest protected-language CE rise. Across the six subcomponent edits, the protected-language rise is 0.0003 to 0.0053 nat/token. The other methods raise it by about 0.03 to 0.88 (Table 2).

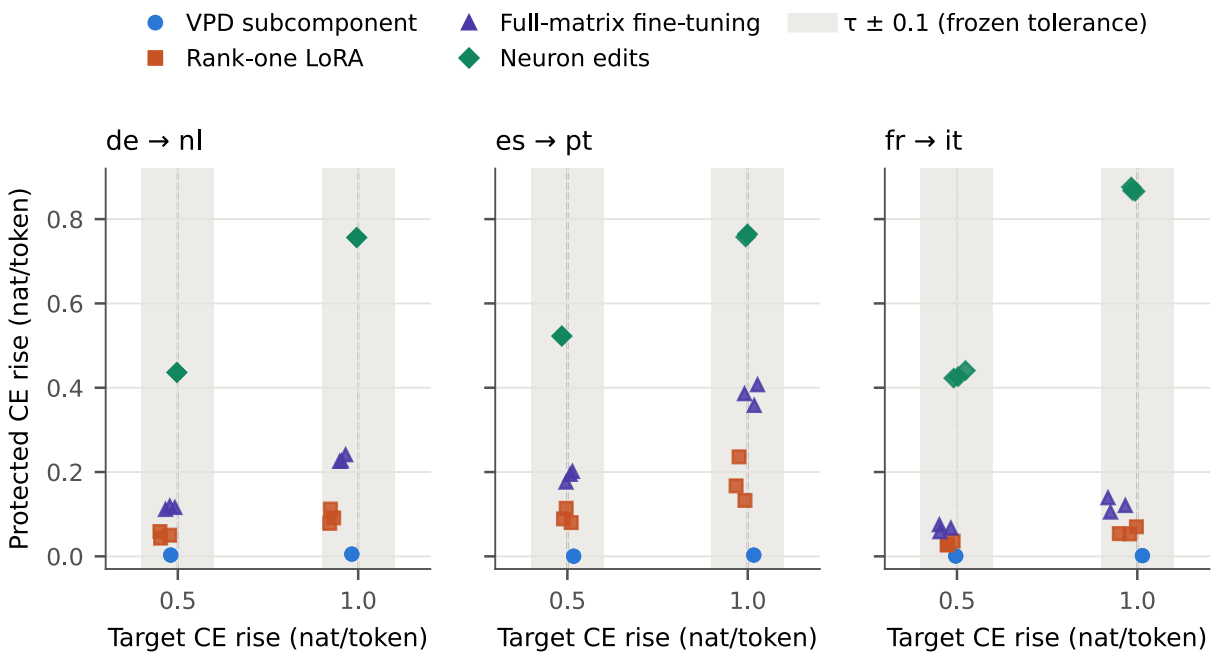


Figure 1: Target CE rise against protected CE rise for every edit on the 58 confirmatory dates (subcomponent edits: one per pair and τ, other methods: one per draw). The bands mark τ ± 0.1, the frozen tolerance of the target flag.

3.2 The protected contrasts

To assess the protected-language differences across session dates, I compare each subcomponent edit directly with each comparator. On the confirmatory dates, all 18 claim contrasts that have an estimate lie below zero, with upper interval ends at −0.023 or below, and every verdict is "reproduced on confirm" (Figure 2, top row, Table 3). The largest difference between a confirmatory estimate and its development estimate is 0.033.

The 18 contrasts share six subcomponent edits: each edit is compared with three comparators. Across draws, the comparators' training samples and the learned updates' random seeds vary; the subcomponent edits stay fixed. The contrasts are therefore dependent.

3.3 Achieved target effects

The edits were calibrated to the same effect on validation text. On the readout dates the achieved effects differ somewhat (Figure 1). To assess whether less protected damage came with a smaller target effect, I estimated, post hoc, the difference in target CE rise between the subcomponent edit and each comparator, its *target gap*, in one simultaneous family with the protected contrasts (method details). A target gap counts as agreement if its whole interval lies within ±0.1 nat/token. This band was chosen after the results. It compares two methods and is separate from the frozen target flag, which compares one edit with τ.

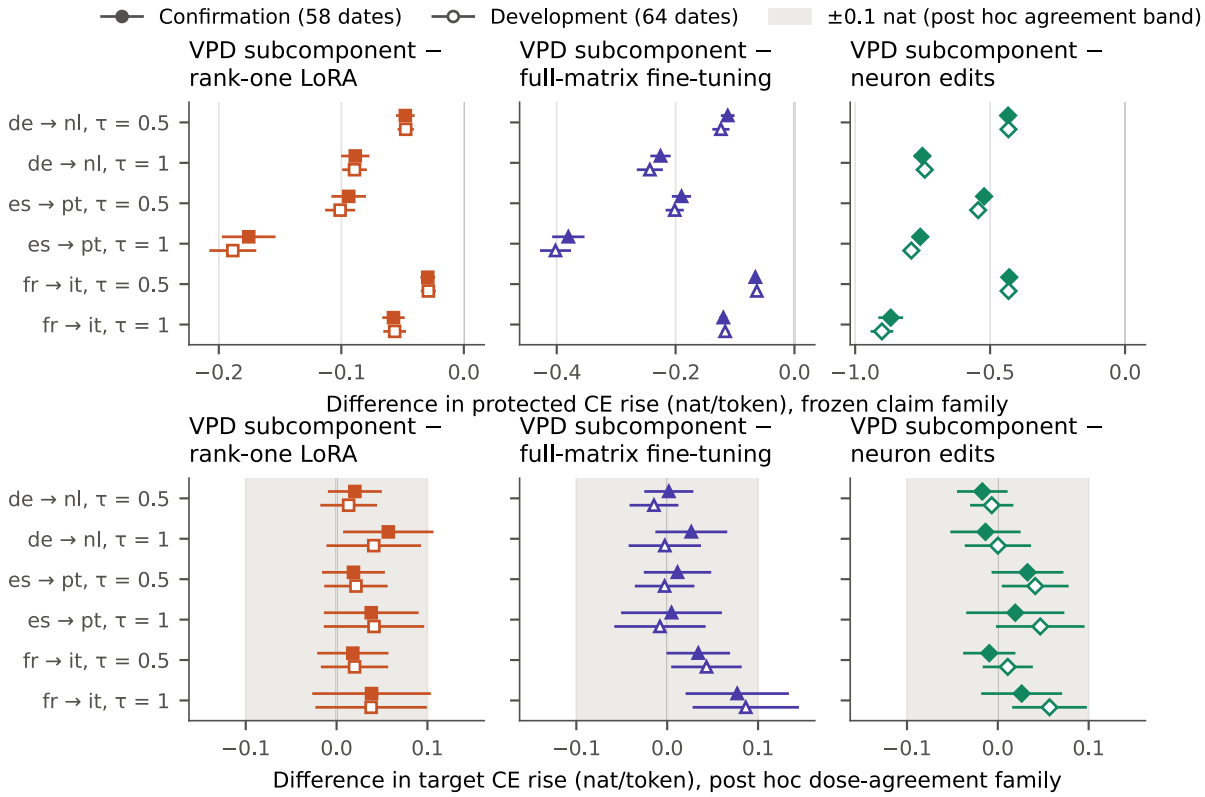


Figure 2: Contrasts of the subcomponent edit with each claim comparator, on the confirmatory (filled, upper marker) and development (open, lower marker) dates, with simultaneous 95% intervals. Top: difference in protected CE rise, the frozen claim family. The axes differ between columns. Bottom: difference in achieved target CE rise, from the post hoc family of Section 3.3, with the ±0.1 nat/token agreement band. Negative values in the top row mean less protected-language damage for the subcomponent edit. Positive values in the bottom row mean a larger target effect.

On the confirmatory dates, 15 of the 18 target gaps agree by this criterion (Figure 2, bottom row). The three exceptions are all at $\tau = 1$, and in each the point estimate lies inside the band:

- German and Dutch against rank-one LoRA: +0.057 [+0.007, +0.107]
- French and Italian against full-matrix fine-tuning: +0.077 [+0.020, +0.134]
- French and Italian against rank-one LoRA: +0.038 [−0.027, +0.104].

On the development dates, 17 of 18 target gaps agree. The exception is French and Italian against full-matrix fine-tuning. All four exceptions have positive point estimates, though the interval for French and Italian against rank-one LoRA permits a smaller target effect for the subcomponent edit. In this larger post hoc family, all 18 protected-CE intervals and all 18 protected-KL intervals lie below zero on both readouts. Protected KL is $D_{\mathrm{KL}}(p_{W_0} \,\|\, p_{\mathrm{edit}})$, averaged over next-token positions on protected-language text.

3.4 English: where the learned updates are better

Less damage to the related language does not imply better preservation of English. The guard rule only asks whether English CE rose past its limit. A finer measure is the KL divergence $D_{\mathrm{KL}}(p_{W_0} \,\|\, p_{\mathrm{edit}})$ of the edited model's English next-token distribution from that of W_0, averaged over positions. By this measure, both learned updates preserve English better than the subcomponent edits (Table 2). On the development readout, over three pairs × three draws × two values of τ × four English sets (the validation set and the three guards), the LoRA update has the lower KL in 72 of 72 comparisons and full-matrix fine-tuning in 65 of 72. On the English confirmatory text both have the lower KL in 18 of 18.

The subcomponent edits also change English little in absolute terms: their English KL on the confirmatory dates is below 10^{-3} nat/token. Their weight changes have larger norms: 0.29 to 0.47, against 0.12 to 0.24 for the

LoRA updates and 0.07 to 0.11 for full-matrix fine-tuning. The learned updates were trained with an explicit English KL term. None of the methods had a term for the protected language. The comparison therefore measures how well the methods spare a related language absent from their training objective.

Table 2: What the edits did on the 58 confirmation dates, as the range over 9 edits per row (3 targets × 3 draws, with the VPD subcomponent draws sharing one edit per target). Rises are CE rises in nat/token on the target, the protected language and English. C: confirmation. D: development. English KL is from the base model. On development data it covers the English validation set and the three guards. Edit norm: Frobenius norm of the weight change.

Method	τ	Target rise	Protected rise	English rise	English KL, C (10^{-3})	English KL, D (10^{-3})	Edit norm
VPD subcomponent	0.5	0.48–0.52	0.0003–0.0030	0.0002–0.0012	0.28–0.41	0.17–0.56	0.29–0.34
	1	0.98–1.02	0.0017–0.0053	0.0003–0.0019	0.49–0.77	0.31–1.01	0.40–0.47
Rank-one LoRA	0.5	0.45–0.51	0.027–0.114	−0.0002 to 0.0001	0.05–0.08	0.04–0.16	0.12–0.17
	1	0.92–1.00	0.053–0.236	−0.0004 to 0.0001	0.08–0.15	0.08–0.25	0.15–0.24
Full-matrix fine-tuning	0.5	0.45–0.51	0.058–0.201	0.0007–0.0014	0.08–0.21	0.08–0.24	0.07–0.08
	1	0.92–1.03	0.105–0.407	0.0010–0.0020	0.16–0.42	0.15–0.49	0.10–0.11
Neuron edits	0.5	0.48–0.52	0.423–0.523	−0.0008 to 0.0010	3.29–5.63	0.61–5.45	1.12–3.16
	1	0.98–1.00	0.756–0.876	0.013–0.038	13.38–38.67	11.85–39.86	5.85–9.80

3.5 The randomized-dictionary control gave no estimate

The randomized-dictionary control was intended to test how selective an edit of the same form is when the subcomponents are unrelated to the model's weights. Applying the same selection rule, it found no feasible subcomponent in any of the 9 target–draw combinations: no shortlisted subcomponent reached even 0.5 nat/token on the target training draw within the dose limit of 2. The randomized-dictionary control therefore has no edit, and its six claim contrasts have no estimate. Because some planned contrasts are empty, the frozen analysis has the overall status "conditional" rather than "complete".

No comparison was possible within the dose limit. Larger doses remain untested, and the alternative selectors do not isolate the dictionary's contribution either (Section 3.6; Section 4).

3.6 Subcomponents chosen by other rules

Four alternatives test other choices within the same dictionary. Two take the first step of orthogonal matching pursuit (Pati et al. 1993; Mallat and Zhang 1993): the subcomponent with the largest absolute correlation with a learned update, using that correlation's sign. One uses the full-matrix fine-tuning update; the other uses a rank-one LoRA across all 24 matrices, unlike the main one-matrix LoRA comparator. A causal-importance heuristic chooses the largest mean target-minus-English importance and a negative sign, using the per-position importance learned in VPD (Bushnaq et al. 2026, sec. 2.3). It adapts Bushnaq's top-16 ranking to one subcomponent (Bushnaq 2026). The fourth reuses the previously chosen German subcomponent.

All alternatives use the same calibration rule. Some failed to reach τ or lacked an eligible learned-update checkpoint, so some contrasts use fewer draws and others have no estimate. Every estimable contrast is at or below zero. The causal-importance heuristic picked `0.attn.o:41` in 8 of 9 target–draw combinations and, on the confirmatory dates, raised protected CE by 0.48 to 1.37 nat/token more than the selected subcomponent, at achieved target rises close to the selected subcomponent's. The previously chosen German subcomponent gives an identical edit. Selectivity therefore varies among subcomponents of the same dictionary.

4 Longer training and alternative rank-one directions

The main comparison leaves open how the result depends on the learned updates' training recipe and on the available rank-one directions. After the confirmatory readout, I examined these dependencies in exploratory comparisons on the previously evaluated development dates. The six subcomponent edits were compared with learned updates trained for 2,000 steps and a whitened selector that searches rank-one directions outside the VPD dictionary using the target gradient and an approximate English Fisher curvature. All were calibrated to the same τ. A separate SVD control applies the selection rule to singular components of the original weight matrices. All 18 protected-CE intervals of the all-run family and all 12 of the family excluding unstable learned runs lie below zero. The achieved target rises differ by at most 0.07 nat/token. The appendix gives the comparison rules and chronology.

The expanded training recipe lowered the protected-language rise of full-matrix fine-tuning in all six target and τ combinations and that of LoRA in five of six. Steps, margin, grid and seeds changed together, with retention still limited to English. Convergence remains unestablished at either budget.

The whitened rank-one selector uses the selected VPD subcomponent's norm and the main selector's ε, leaving the dictionary's contribution entangled with the selector's. The SVD comparator gave no estimate: every edit that reached τ raised an English guard by more than 0.1 nat/token, and three of nine did not reach $\tau = 1$.

5 Limits of these conclusions

The result covers six edits in one four-layer model, three language pairs and two values of τ. The measured outcomes are protected-language CE on parliamentary text and English CE guards and KL. The confirmatory readout tests fixed edits on new text in the same languages; uncertainty varies session dates only (Section 2.5). Calibration matches mean target effects, while the learned updates optimize a per-window hinge. English retention is not matched across methods. Testing the selection rule in other models, explaining the edits' selectivity and evaluating tasks beyond next-token prediction remain open.

6 Conclusion

Across three language pairs in this model, six subcomponent edits raised related-language CE by at most 0.0053 nat/token on 58 new session dates. All 18 simultaneous 95% intervals for the differences from the three comparators lie below zero. Target effects were matched at 0.5 or 1 nat/token on validation text; in a post hoc check on evaluation text, 15 of the 18 target-gap intervals lay wholly within ± 0.1 nat/token. The three remaining mean gaps were positive, though one interval also allowed a smaller subcomponent effect. Learned updates preserved English output better.

Exploratory comparisons on previously evaluated development dates also favoured subcomponent edits for related-language damage against a whitened rank-one selector and learned updates trained for 2,000 steps. These results concern fixed edits: prior protected-language access (Section 2.4) and the inconclusive randomized-dictionary control leave the selection rule's generalization and the dictionary's contribution unresolved.

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References

Braun, Dan, Lucius Bushnaq, Stefan Heimersheim, Jake Mendel, and Lee Sharkey. 2025. *Interpretability in Parameter Space: Minimizing Mechanistic Description Length with Attribution-Based Parameter Decomposition*. https://doi.org/10.48550/arXiv.2501.14926.

Bushnaq, Lucius. 2026. *Exploration: Fine-Tuning with Parameter Decomposition*. LessWrong, 25 June 2026. https://www.lesswrong.com/posts/ieoWstubDQWLrMnhH/exploration-fine-tuning-with-parameter-decomposition.

Bushnaq, Lucius, Dan Braun, Oliver Clive-Griffin, et al. 2026. *Interpreting Language Model Parameters*. Goodfire and MATS. https://www.goodfire.com/research/interpreting-lm-parameters.

Bushnaq, Lucius, Dan Braun, and Lee Sharkey. 2025. *Stochastic Parameter Decomposition*. https://doi.org/10.48550/arXiv.2506.20790.

Efron, Bradley. 1979. "Bootstrap Methods: Another Look at the Jackknife." *The Annals of Statistics* 7 (1): 1–26. https://doi.org/10.1214/aos/1176344552.

Federmann, Christian, Tom Kocmi, and Ying Xin. 2022. "NTREX-128 – News Test References for MT Evaluation of 128 Languages." *Proceedings of the First Workshop on Scaling up Multilingual Evaluation* (Online), 21–24. https://doi.org/10.18653/v1/2022.sumeval-1.4.

Gao, Leo, Stella Biderman, Sid Black, et al. 2020. *The Pile: An 800GB Dataset of Diverse Text for Language Modeling*. https://doi.org/10.48550/arXiv.2101.00027.

Hu, Edward J., Yelong Shen, Phillip Wallis, et al. 2022. "LoRA: Low-Rank Adaptation of Large Language Models." *International Conference on Learning Representations*. https://openreview.net/forum?id=nZeVKeeFYf9.

Kassis, Timothy, Vinayak Agarwal, Yuhuan He, Darshil Patel, and Aubrey M. Brueckner. 2026. *Scientific Agent Skills: A Library of Procedural Knowledge for Research Agents*. https://doi.org/10.48550/arXiv.2609.00065.

Kingma, Diederik P., and Jimmy Ba. 2015. "Adam: A Method for Stochastic Optimization." *International Conference on Learning Representations*. https://arxiv.org/abs/1412.6980.

Koehn, Philipp. 2005. "Europarl: A Parallel Corpus for Statistical Machine Translation." *Proceedings of Machine Translation Summit X: Papers* (Phuket, Thailand), 79–86. https://aclanthology.org/2005.mtsummit-papers.11/.

Kojima, Takeshi, Itsuki Okimura, Yusuke Iwasawa, Hitomi Yanaka, and Yutaka Matsuo. 2024. "On the Multilingual Ability of Decoder-Based Pre-Trained Language Models: Finding and Controlling Language-Specific Neurons." *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)* (Mexico City, Mexico), 6919–71. https://doi.org/10.18653/v1/2024.naacl-long.384.

Kunstner, Frederik, Philipp Hennig, and Lukas Balles. 2019. "Limitations of the Empirical Fisher Approximation for Natural Gradient Descent." *Advances in Neural Information Processing Systems* 32: 4158–69. https://proceedings.neurips.cc/paper/2019/hash/46a558d97954d0692411c861cf78ef79-Abstract.html.

Mallat, Stéphane G., and Zhifeng Zhang. 1993. "Matching Pursuits with Time-Frequency Dictionaries." *IEEE Transactions on Signal Processing* 41 (12): 3397–415. https://doi.org/10.1109/78.258082.

Martens, James. 2020. "New Insights and Perspectives on the Natural Gradient Method." *Journal of Machine Learning Research* 21 (146): 1–76. https://jmlr.org/papers/v21/17-678.html.

Martens, James, and Roger Grosse. 2015. "Optimizing Neural Networks with Kronecker-Factored Approximate Curvature." *Proceedings of the 32nd International Conference on Machine Learning* 37: 2408–17. https://proceedings.mlr.press/v37/martens15.html.

Meng, Kevin, David Bau, Alex Andonian, and Yonatan Belinkov. 2022. "Locating and Editing Factual Associations in GPT." *Advances in Neural Information Processing Systems* 35: 17359–72. https://doi.org/10.52202/068431-1262.

Merity, Stephen, Caiming Xiong, James Bradbury, and Richard Socher. 2017. "Pointer Sentinel Mixture Models." *International Conference on Learning Representations*. https://arxiv.org/abs/1609.07843.

Montiel Olea, José Luis, and Mikkel Plagborg-Møller. 2019. "Simultaneous Confidence Bands: Theory, Implementation, and an Application to SVARs." *Journal of Applied Econometrics* 34 (1): 1–17. https://doi.org/10.1002/jae.2656.

Pati, Y. C., R. Rezaiifar, and P. S. Krishnaprasad. 1993. "Orthogonal Matching Pursuit: Recursive Function Approximation with Applications to Wavelet Decomposition." *Proceedings of 27th Asilomar Conference on Signals, Systems and Computers* 1: 40–44. https://doi.org/10.1109/ACSSC.1993.342465.

Tang, Tianyi, Wenyang Luo, Haoyang Huang, et al. 2024. "Language-Specific Neurons: The Key to Multilingual Capabilities in Large Language Models." *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (Bangkok, Thailand), 5701–15. https://doi.org/10.18653/v1/2024.acl-long.309.

Yao, Yunzhi, Peng Wang, Bozhong Tian, et al. 2023. "Editing Large Language Models: Problems, Methods, and

Opportunities." *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing* (Singapore), 10222–40. https://doi.org/10.18653/v1/2023.emnlp-main.632.

Zhang, Zhihao, Jun Zhao, Qi Zhang, Tao Gui, and Xuanjing Huang. 2024. "Unveiling Linguistic Regions in Large Language Models." *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (Bangkok, Thailand), 6228–47. https://doi.org/10.18653/v1/2024.acl-long.338.

A Method details

Text. Europarl v7 (Koehn 2005) supplies one 257-token window per session date (256 predictions). Each target has three draws of 64 of its 128 training windows and 32 validation windows for calibration. The English validation set has 64 windows: 32 Europarl and 32 English-filtered Pile. The retain pool has 256 windows: 128 Europarl English training, 64 WikiText-2 validation and 64 English-filtered Pile validation.

Selection. For softmax cross-entropy, the model Fisher equals the generalized Gauss–Newton matrix (Martens 2020, sec. 9.2), unlike the empirical Fisher based on data labels (Kunstner et al. 2019, sec. 1). The estimate q_c averages squared directional scores using eight model-sampled label sets per retain window. The floor ε is the 1st percentile of $q_c/\|A_c\|^2$. The top 64 subcomponents are re-ranked using the larger of q_c and a second estimate on 256 other English windows, before the dose search of Section 2.1. Curvature checks passed on the NVIDIA L40S used for the original runs, but the batch-splitting check failed on the tested A10 and L4. Verified selector reruns cover only the German target's first draw on the L40S, with the VPD dictionary and one randomized dictionary. GPU-dependent Fisher-label sampling can change selections on other hardware.

Calibration. The first dose-grid crossing of τ is bracketed, then bisected for at most five steps until within the matching tolerance. Neuron edits scale the top k neurons' output weights by $1 - f$: calibration takes the smallest power-of-two k from 1 to 256 whose full removal reaches τ, then calibrates f. Randomized dictionaries use a separate seed per draw.

Learned updates. The 200-step updates use Adam (Kingma and Ba 2015), 10 warm-up steps, gradient clipping at 1.0, and 8 target and 8 English windows per step. Calibration scales the final update D to βD, with $\beta \leq 1$. The learning rate is chosen once per target on draw 0, by the lowest English validation KL at $\tau = 1$. Every target chose 3e-3 for LoRA and 1e-6 for full-matrix fine-tuning, respectively the upper and lower edges of the tested grids.

A.1 Extended comparisons

The extended learned updates use 2,000 steps, a hinge margin of 1.5, wider learning-rate grids and three seeds per draw. The whitened selector maximizes the squared target gradient relative to damped K-FAC curvature on English text (Martens and Grosse 2015), then tests doses from 1/64 to 8. The SVD control applies the main selection rule unchanged to the singular components of all 24 weight matrices; these reconstruct the weights exactly, whereas VPD leaves a residual. The exploratory rerun selected the same VPD subcomponents in all nine target–draw combinations.

Contrasts compare unpaired means over each method's matched and usable runs in one simultaneous family. A second family excludes unstable learned runs: 7 of 27 LoRA runs had English validation KL at step 2,000 greater than twice their minimum from step 100 onward. Training diagnostics did not establish objective convergence and were not used to select edits.

The comparisons were specified before the runs, on 12 October 2026. A 5% tolerance for reproducing the curvature floor across GPU models was added after one selection finished, before its result was read. Analysis rules were fixed before readout, after selection outcomes, learning-rate choices and the rate grid's target and English validation effects were known. The analyzer's interval checks were implemented after readout but before its measurements were opened or analysed. This chronology comes from the author's records and is not independently timestamped.

A.2 Readouts and intervals

Readouts. Development guards use Europarl English test text (64 windows), WikiText-2 test text (32) and Pile test documents (512). The confirmatory guard uses English from the 58 confirmatory dates, comprising 48 main and 10 reserve dates analysed together.

Intervals. The bootstrap uses 2,000 date resamples with replacement, shared across each family. A contrast's standard error is the standard deviation of its resampled means. Its interval is the estimate $\pm\, c$ standard errors, where c is the 95% quantile of the largest absolute standardized deviation in the family.

Verdicts. The first applicable rule determines the result: missing intervals or technical failures give "not established"; a target flag in any used draw gives "effect not maintained"; an interval excluding zero with the development sign gives "reproduced on confirm", or with the opposite sign "not reproduced". All other cases are "not established".

Target gaps. The post hoc family has 54 coordinates per readout: target gaps, protected-CE contrasts and protected-KL contrasts. Per-date values are averaged over draws, with 20,000 shared date resamples.

A.3 Protocol records and prior access

- **Freeze and readout.** The selection, comparison and analysis rules were frozen on 11 October 2026, immediately before the development readout. Later that day, the unchanged edits were evaluated once on 58 further dates, under an amendment fixed before any of those windows was parsed.
- **Prior knowledge.** German subcomponent `0.attn.o:268` had been chosen with Dutch in view in an earlier experiment and was the top German candidate in a pilot of this selector. Dutch, Portuguese and Italian development text and NTREX Dutch and Italian text (Federmann et al. 2022) had been read or used previously. The 64 development dates therefore do not provide a fresh test.
- **Pretraining.** The Pile contains a European Parliament component (Gao et al. 2020, sec. 2.17); whether the model's training subset includes it is unknown. Dates new to this workflow may have appeared in pretraining.
- **Record limits.** The chronology relies on the author's documents and run log, without independent timestamps. The original protocol, freeze record and frozen code list are unpublished. Their hashes identify contents but do not establish dates, so there is no timestamped preregistration.

B The claim contrasts

Table 3 lists the 18 claim contrasts of Figure 2 with their intervals. The six randomized-dictionary contrasts have no estimate (Section 3.5).

Table 3: The 18 claim contrasts, VPD subcomponent minus comparator, in nat/token, with simultaneous 95% intervals. Protected: difference in protected CE rise, the frozen claim family. Target gap: difference in achieved target CE rise, from the post hoc 54-coordinate family.

Pair	τ	Comparator	Protected, confirmation	Protected, development	Target gap, confirmation
de → nl	0.5	Rank-one LoRA	−0.048 [−0.055, −0.040]	−0.047 [−0.054, −0.041]	+0.020 [−0.010, +0.050]
		Full-matrix fine-tuning	−0.112 [−0.124, −0.101]	−0.124 [−0.138, −0.109]	+0.002 [−0.025, +0.029]
		Neuron edits	−0.433 [−0.457, −0.410]	−0.432 [−0.456, −0.408]	−0.017 [−0.045, +0.011]
	1	Rank-one LoRA	−0.089 [−0.100, −0.077]	−0.089 [−0.099, −0.079]	+0.057 [+0.007, +0.107]
		Full-matrix fine-tuning	−0.225 [−0.242, −0.208]	−0.243 [−0.265, −0.221]	+0.027 [−0.013, +0.066]
		Neuron edits	−0.751 [−0.782, −0.720]	−0.742 [−0.772, −0.712]	−0.014 [−0.052, +0.025]
es → pt	0.5	Rank-one LoRA	−0.094 [−0.108, −0.080]	−0.101 [−0.113, −0.089]	+0.019 [−0.016, +0.053]
		Full-matrix fine-tuning	−0.190 [−0.206, −0.174]	−0.202 [−0.217, −0.186]	+0.011 [−0.026, +0.048]
		Neuron edits	−0.522 [−0.549, −0.496]	−0.544 [−0.570, −0.519]	+0.033 [−0.007, +0.072]
	1	Rank-one LoRA	−0.176 [−0.197, −0.154]	−0.189 [−0.208, −0.169]	+0.038 [−0.014, +0.090]
		Full-matrix fine-tuning	−0.380 [−0.408, −0.353]	−0.402 [−0.428, −0.376]	+0.005 [−0.051, +0.060]
		Neuron edits	−0.759 [−0.790, −0.728]	−0.792 [−0.821, −0.763]	+0.019 [−0.035, +0.073]
fr → it	0.5	Rank-one LoRA	−0.029 [−0.035, −0.023]	−0.029 [−0.035, −0.023]	+0.018 [−0.021, +0.057]
		Full-matrix fine-tuning	−0.066 [−0.071, −0.061]	−0.063 [−0.068, −0.058]	+0.034 [−0.001, +0.069]
		Neuron edits	−0.429 [−0.456, −0.402]	−0.432 [−0.456, −0.407]	−0.009 [−0.038, +0.019]
	1	Rank-one LoRA	−0.057 [−0.067, −0.048]	−0.056 [−0.066, −0.047]	+0.038 [−0.027, +0.104]
		Full-matrix fine-tuning	−0.120 [−0.126, −0.113]	−0.117 [−0.123, −0.110]	+0.077 [+0.020, +0.134]
		Neuron edits	−0.868 [−0.914, −0.822]	−0.901 [−0.943, −0.859]	+0.026 [−0.018, +0.071]

Reproducibility and availability. Code, saved measurements and reproduction instructions are on GitHub. Saved per-date measurements support analysis replay. The confirmatory token windows and texts are not released, so that model evaluation cannot be rerun from this checkout. The guide describes the available model procedures.